

Cover letter — NBDT submission

Dear Editors of *Neurons, Behavior, Data analysis, and Theory*,

We submit our manuscript, “**Neural-Grounded Computational Decision Phenotypes: Honest-by-Construction Prediction of Choice, with Provable Abstention**”, for consideration at NBDT.

What the paper does. We build a model that predicts a person’s decisions from a small set of neuroeconomic parameters, grounded in Knutson’s Affect-Integration-Motivation account, and we make it *honest by construction*: every prediction carries a distribution-free coverage guarantee and a strictly-proper score, and the model abstains when the data cannot support a prediction. We validate the estimator on real human choices from two public mixed-gambles fMRI datasets, ground the affective channel in real fMRI and EEG, and — where a direct within-subject neural-to-behavior link is underpowered — the model abstains and a correlation-transitivity argument shows the grounding does not require it.

Why NBDT. The work is exactly at the intersection NBDT serves: quantitative models of behavior, grounded in neural data, with careful data analysis and explicit theory about what the data can and cannot support. Our central methodological stance — reporting simulation results as positive/negative controls and real-data results as measured effects, and treating abstention as a first-class output — fits NBDT’s emphasis on rigorous, transparent computational neuroscience.

Openness and reproducibility. Every number traces to a committed analysis script and a result file. All datasets are public (NARPS ds001734; Tom et al. 2007 ds000005; EEG ds003458). We are glad to release the full code repository on acceptance.

Honesty about scope. We are explicit throughout about what is established on real data (the estimator, honesty layer, cross-dataset transfer, individuation, and two-modality neural grounding of the affective channel) versus what is shown in simulation (the confounded-target abstention and the attribution-scale extension) versus what remains underpowered and is therefore abstained (the direct individual neural-to-behavior correlation). No individual is ever scanned, and we make no claim to measure any specific person’s brain.

We believe this is a good fit for NBDT and look forward to the reviewers’ feedback.

Sincerely,

Aayush Gandhi